SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I Eddu Suchier/19524070 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Scenario-based

4. Public Key Cryptography for Email Security

A. Describe how RSA can be used for encryption and decryption in this scenario.

RSA is a public key cryptography algorithm that provides secure email communication by using two different keys: a public key and a private key. Every user has a unique key pair, where the public key is shared openly, and the private key is kept secret. When a sender wants to send a confidential email, they encrypt the message using the recipient's public key. Once encrypted, the email can only be decrypted using the recipient's private key, ensuring that no unauthorized person can read the message during transmission. In addition to encryption, RSA is also used for digital signatures. The sender signs the email using their private key, and the recipient verifies the signature using the sender's public key. This process confirms the sender's identity, ensures the message has not been modified, and provides confidentiality, authentication, integrity, and non-repudiation in email communication.

B. Explain how key distribution and management play a role in maintaining security.

Key distribution and management are critical for maintaining the security and reliability of RSA-based email systems. Public keys must be distributed securely so that users can be confident they are communicating with the intended recipient rather than an attacker. This is commonly achieved through digital certificates issued by trusted Certificate Authorities (CAs), which verify the ownership of public keys. At the same time, private keys must remain confidential and be stored securely using encryption, hardware security modules, or password-protected key stores. Organizations should also establish proper key management practices, including regular key generation, secure backup, periodic key renewal, certificate updates, and immediate revocation of compromised keys. These measures help prevent unauthorized access, impersonation, and data breaches while ensuring that secure email communication remains confidential, authentic, and trustworthy over time.

5. Cryptography for IoT Devices

A. Analyze the constraints of IoT devices.

IoT devices are designed to operate with limited hardware resources, making security implementation more challenging than in traditional computing systems. Most IoT devices have low processing power, limited memory, and restricted storage capacity, which makes it difficult to run computationally intensive cryptographic algorithms. They are also often battery-powered, so energy efficiency is an important consideration because complex encryption methods can reduce battery life. In addition, IoT devices usually communicate over low-bandwidth wireless networks, requiring cryptographic protocols that minimize communication overhead and data transmission. Many IoT devices are deployed in remote or unattended environments, increasing the risk of physical tampering and cyberattacks. Therefore, the chosen cryptographic solution must provide strong security while remaining lightweight, efficient, and suitable for devices with limited computational and energy resources.

B. Recommend a suitable cryptographic approach and justify your answer.

Elliptic Curve Cryptography (ECC) is the most suitable cryptographic approach for IoT devices because it provides strong security with much smaller key sizes than RSA. For example, a 256-bit ECC key offers a security level comparable to a 3072-bit RSA key. The smaller key size reduces computation time, memory usage, storage requirements, and network bandwidth, making ECC highly efficient for low-power and resource-constrained IoT devices. It also consumes less energy, which helps extend the battery life of devices that operate continuously in the field. While RSA is widely used and well-established, its larger key sizes require more processing power and memory, making it less practical for most IoT environments. Therefore, ECC is the preferred choice for securing smart home IoT devices because it achieves a better balance between security, efficiency, and performance while meeting the resource limitations of embedded systems.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand